Exploring Heterogeneous Feature Representation for Document Layout Understanding

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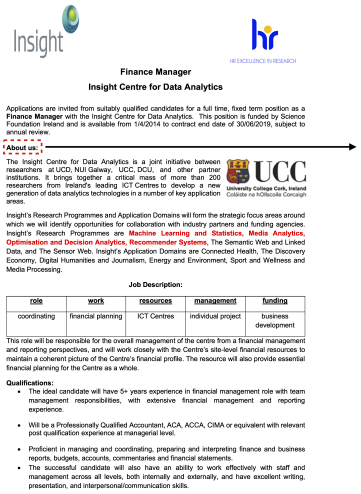
***Abstract*—There are increasing interests in document layout representation learning and understanding. Transformer, with its great power, has become the mainstream model architecture and achieved promising results in this area. As elements in a document layout consist of multi-modal and multi-dimensional features such as position, size, and its text content, prior works represent each element by summing all feature embeddings into one unified vector in the input layer, which is then fed into the self-attention for element-wise interaction. However, this simple summation would potentially raise mixed correlations among heterogeneous features and bring noise to the representation learning. In this paper, we propose a novel two-step disentangled attention mechanism to allow more flexible feature interactions in the self-attention. Furthermore, inspired by the principles of document design (e.g., contrast, proximity), we propose an unsupervised learning objective to constrain the layout representations. We verify our approach on two layout understanding tasks, namely element role labeling and image captioning. Experiment results show that our approach achieves state-of-the-art performances. Moreover, we conduct extensive studies and observe better interpretability using our approach.**

***Index Terms*—Layout Representation Learning, Layout Understanding, Attention Disentanglement**

I. Introduction

Document layout representation and understanding are crucial parts in document intelligence, which serve as fundamental blocks to support many downstream applications such as document analysis [1] and retrieval [2]. Advanced models, such as LAMBERT [3], LayoutLM [4], [5] and DocFormer [6], have been proposed to integrate multi-modality information (textual, spatial and visual) for digital document analysis (e.g., forms and receipts recognition). Meanwhile, CanvasEmb [7] focuses on only the layout modality by considering design-related features such as type, position and font to construct document representation and enables applications in document design. Most of these works adopt Transformer [8] as the backbone encoder, which is a powerful architecture for its capability of contextualization and generalization.

Taking a sequence of tokens with multiple features as input, the common approach in the Transformer-based models is to directly sum different feature embeddings of a token into one vector in the input layer. For example in modeling text sequences, each word is represented by the sum of its word embedding and position embedding. The token embeddings are then fed into the self-attention layers for token-wise contextualization. However, the simple summation of feature embeddings for a token could bring mixed correlations between features in the self-attention, especially when the number of features is large. Several works of disentangled attention [6], [9], [10] have been proposed to alleviate this issue. More specifically, they expand the dot-product function of keys and queries in the self-attention, and heuristically drop a subset of key-query pairs such as word-to-position, position-to-word as they think such correlations are noisy. As for document layouts, one element contains many features as an example shown in Fig. 1. It would be difficult to enumerate and filter unnecessary key-query feature pairs only with predefined heuristics.



Property	Value
Element ID	4
Word Count	2
Bold	TRUE
Italic	FALSE
Underline	FALSE
Allcaps	FALSE
Alignment Type	Left
Indentation	0.0
FontSize	22
Color	[255, 0, 0, 0]
FontName	Calibri
Bulleted List	FALSE

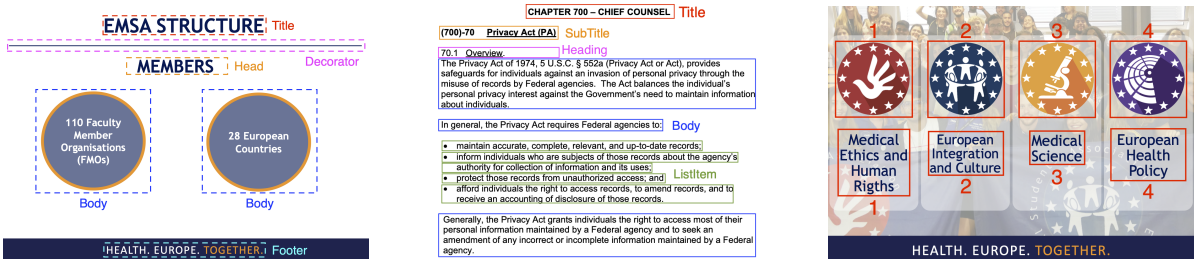
Fig. 1: An example of a word document layout. Each element (defined as paragraph here) is represented with multiple design-related features.

In this paper, we follow the work of CanvasEmb document layout representation learning (without considering the actual content), and propose a more flexible attention disentanglement for features interaction. Instead of simple summation of heterogeneous features embeddings, we propose a two-step

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(a) element roles in slide (b) element roles in word document (c) image captioning in slide

Fig. 2: Two layout understanding tasks. (a)(b) Element role labeling in slide and word document, with each element assigned to a semantic role. (c) Image captioning with four image-caption pairs.

disentangled attention mechanism (DeAttnFormer): (1) intra-feature attention; we preserve separate feature embeddings for each element in a layout, and conduct element-wise self-attention respective to each feature channel. (2) inter-feature attention; within an element, we model feature-to-feature correlations by updating the feature embeddings with each other as contexts. In this way, the model has the flexibility to fully capture contexts of elements, as well as automatically learn the importance of different pairwise feature correlations.

Further, we propose an unsupervised learning objective to constrain layout representation. Layouts usually follow certain design principles, such as contrast and proximity. For example, the principle of contrast emphasizes to create strong visual disparities (e.g., color, font size) between elements of different types. Inspired by this, we design an auxiliary loss to constrain the element representations to be closer when they share more similar layout settings. This requires no annotation effort for training, and we show its effectiveness in experiments when coupled with different models.

We verify our approach on two layout understanding tasks: (1) element role labeling, which classifies each element into a semantic role (e.g. *title*, *body*); (2) image captioning, which detects if an image and a text box belongs to an image-captioning pair. In addition to the slide data proposed in CanvasEmb, we collect a dataset of 2,609 word documents from the Web with human-annotated element role labels. Experiments on both datasets show that our approach consistently performs better than the baselines. The constrained loss further boosts model performance. Through in-depth analysis, we show that our approach obatains more explanatory attention patterns and more reasonable layout representations.

To summarize, the main contributions of this paper are:

- We propose a two-step disentangled attention mechanism to model the complex correlations among heterogeneous features in layout representation learning.
- We design an unsupervised learning objective, which softly constrains the element representations to follow certain design principles without the necessity of annotation.
- We achieve currently the best performances on two layout understanding tasks, and conduct extensive studies of analyzing internal model behaviors.

II. Layout in Documents

Before we introduce our approach, this section first briefly introduces the scope of layout definition and the tasks considered in this paper.

A. Scope of Layout

Layout refers to the arrangement of elements in a document. Layout settings can be varied among different document types (e.g., slide, word document and web page), but the elements share some common features: (1) type features (e.g., whether the element is a textbox, image, or button); (2) geometry features, such as position and rotation; (3) color features, including RGBA and gradient; (4) content-related features but without the actual content, such as text font setting.

An example of a word document is shown in Fig. 1, in which each element (defined as paragraph) contains multiple features that are either categorical (e.g., `AlignmentType`, `FontName`) or numerical (e.g., `ElementID`, `FontSize`). With the set of heterogeneous features, document layouts can be very diverse and complex.

B. Layout Understanding

Layout understanding relates to modeling elements and their relations. This paper considers two fundamental tasks as proposed in CanvasEmb:

- **Element Role Labeling.** This task aims to identify the semantic role of each element, such as *title*, *subtitle* and *footnote*. The definitions of semantic roles are specific to different document types. We show one example for slide in Fig. 2a and one for word document in Fig. 2b.
- **Image Captioning.** Given a pair of an image and a text box in a layout, the goal is to detect if the two elements belong to an image-captioning pair or not. An example is shown in Fig. 2c.

III. Approach

In this section, we present our proposed Disentangled Attention Transformer (DeAttnFormer) for layout representation. Furthermore, an unsupervised learning constraint is introduced based on the document design principles.

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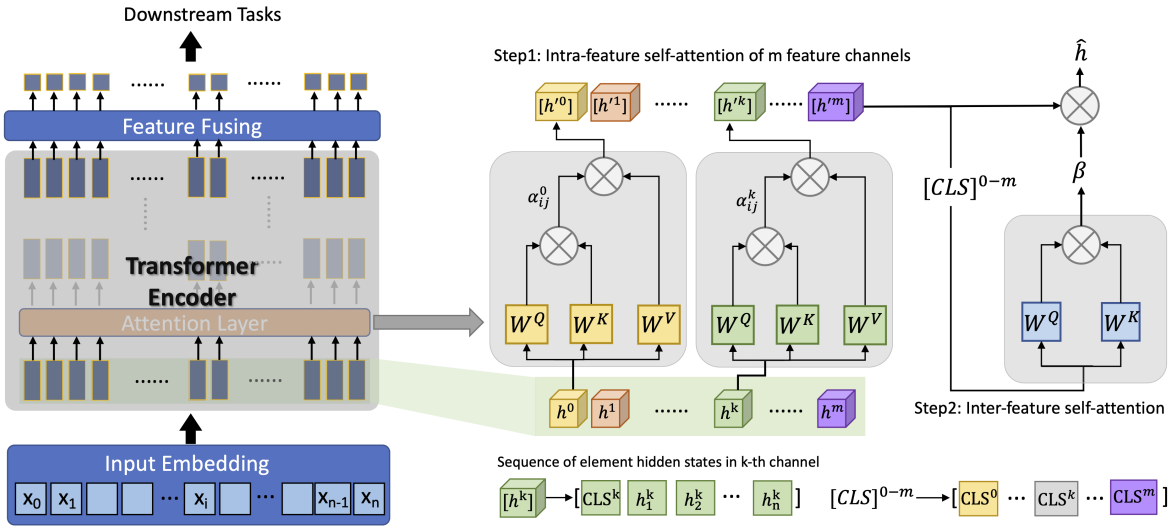


Fig. 3: Overall architecture of our approach. Each element x_i contains m features. $[\mathbf{h}^k]$ refers to the sequence of element embeddings with respect to the k-th feature (superscript indicates feature index while subscripts indicates element index in the sequence). Cubes in different colors represent different feature channels.

A. Transformer with Heterogeneous Features

CanvasEmb is a Transformer-based model designed for layout representation which currently achieves state-of-the-art in several layout understanding tasks. Here we briefly describe the model.

Input. A layout can be viewed as an unordered sequence of n elements $\mathbf{x} = [\mathbf{x}_0, \mathbf{x}_1, \cdots, \mathbf{x}_n]$. Here, $\mathbf{x}_0$ is the special symbol `[CLS]` defined in [11] summarizing and representing the whole sequence. Each element consists of m features denoted as $\mathbf{x}_i = [f_i^0, f_i^1, \cdots, f_i^m]$. Features are categorized into either categorical (e.g., `AlignmentType`) or numerical (e.g., `Indentation`) types with different dimensions and scales.

Element Embedding. For an element, CanvasEmb calculates the embeddings of its features $[\epsilon_i^0, \epsilon_i^1, \cdots, \epsilon_i^m]$. Specifically, the embeddings for categorical features are obtained using a learnable look-up table, while for the numerical features, we adopt the sinusoidal positional encoding [8]. The element embedding can then be calculated as the concatenation of all its feature embeddings with a transformation Θ (equally to weighted summation):

$$\mathbf{h}_i = \Theta[\epsilon_i^0 \oplus \epsilon_i^1 \oplus \cdots \oplus \epsilon_i^m] \tag{1}$$

Self-Attention. The element embeddings are fed into the multi-head self-attention layers for capturing element contexts. Specifically for one head, the attention score can be calculated as:

$$\alpha_{i,j} = \frac{1}{\sqrt{d_K}}(\mathbf{W}_Q\mathbf{h}_i)(\mathbf{W}_K\mathbf{h}_j)^T \tag{2}$$

and the hidden representation $\mathbf{h}_i$ is updated as:

$$\mathbf{h}_i^{'} = \sum_{j=1}^{n} \mathrm{softmax}(\alpha_{i,j})(\mathbf{W}_V\mathbf{h}_j) \tag{3}$$

where W_Q, W_K, W_V are learnable parameters, and d_K is the dimension of W_K.

Issue: Mixed feature correlations. As observed in [9], [10], the simple summation of features embeddings into one vector brings mixed correlations between different information sources, and thus results in unnecessary randomness in the attention. Take an input with two feature embeddings $\{\epsilon^1, \epsilon^2\}$ as an example, the attention in Equation 2 can be expanded as:

$$\alpha_{i,j} \propto \frac{1}{\sqrt{d_K}}(\mathbf{W}_Q(\epsilon_i^1 + \epsilon_i^2))(\mathbf{W}_K(\epsilon_j^1 + \epsilon_j^2))^T \tag{4}$$

which can be further expanded into four key-query terms related to: (1) $\epsilon_i^1\epsilon_j^{1T}$; (2) $\epsilon_i^1\epsilon_j^{2T}$; (3) $\epsilon_i^2\epsilon_j^{1T}$; (4) $\epsilon_i^2\epsilon_j^{2T}$. As the two features represent concepts in different dimensions, it might be noisy to consider the 2nd and 3rd terms as they encode correlations between heterogeneous features. As a solution, previous works decouple the attention by selecting only a subset of the four attention terms with heuristics[1].

In our scenario of document layouts, an element usually contains $m > 2$ features, and there would be m^2 key-query terms in the expanded attention. The effect of correlations between different feature pairs is remained unknown. Heuristically selecting a subset of pairs as in the previous approaches is also challenging.

B. Disentangled Attention

In this paper, we propose a two-step attention mechanism to allow more flexibility in modeling feature correlations.

Step 1: (Element-to-Element) Intra-Feature Attention. Each element is represented using m feature embeddings which are separately sent to the self-attention module. With

[1]For example, DeBERTa removes the position-to-position term by assuming it does not provide useful information.

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respect to the k-th feature channel, the element representation is updated using other elements as contexts:

$$\mathbf{h}_i^{'k} = \sum_{j=1}^{n} \mathrm{softmax}(\alpha_{\mathrm{ij}}^{\mathrm{k}})(\mathbf{W}_{\mathrm{V}}^{\mathrm{k}}\mathbf{h}_{\mathrm{j}}^{\mathrm{k}})$$
$$\alpha_{ij}^{k} = \frac{1}{\sqrt{d_K}}(\mathbf{W}_Q^k\mathbf{h}_i^k)(\mathbf{W}_K^k\mathbf{h}_j^k)^T \quad (5)$$

where $\mathbf{h}_i^k$ represents the i-th element hidden state related to the k-th feature. Query, key, value matrices ($\mathbf{W}_Q^k$, $\mathbf{W}_k^k$, $\mathbf{W}_V^k$) are assigned separately for each feature channel. Note that $[\mathbf{h}_i^{'0}, \cdots, \mathbf{h}_i^{'m}]$ can be computed in parallel for more efficient training especially when the number of features increases.

Step 2: (Feature-to-Feature) Inter-Feature Attention. Some feature pairs might be highly correlated while others are not. For example, in a word document, paragraphs with smaller `ElementID` (e.g., top paragraphs such as title and subtitle) might have larger `FontSize`. To enable the model to automatically learn useful feature correlations, we update the element hidden state $\mathbf{h}_i^{'k}$ in the k-th feature channel with the states in other channels.

We first calculate the weight β^{kq} of the k-th feature attending to the q-th feature as:

$$\beta^{kq} = \frac{1}{\sqrt{d_K}}(\mathbf{W}_Q^k[\mathtt{CLS}]^k)(\mathbf{W}_K^q[\mathtt{CLS}]^q)^T \quad (6)$$

here β^{kq} corresponds to attention weight of the sequence representations $[\mathtt{CLS}]^k$ and $[\mathtt{CLS}]^q$ from two feature channels indexed as k and q, and is computed similar to Equation 2. Within an element, the hidden state $\mathbf{h}_i^{'k}$ in the k-th channel is further updated by taking its hidden states in other feature channels as contexts:

$$\hat{\mathbf{h}}_i^k = \sum_{q=0}^{m} \mathrm{softmax}(\beta^{\mathrm{kq}})\mathbf{h}_{\mathrm{i}}^{'\mathrm{q}} \quad (7)$$

Note that we do not multiply the value matrix here with the hidden vector $\mathbf{h}_i^{'q}$ as computed in step 1, as to avoid redundant transformations. Overall we expect the model to capture element-to-element interactions per feature channel as well as feature-to-feature correlations in each self-attention layer.

Feature Fusing. Finally on the top of the model, we aggregate the hidden states of total m feature channels using the aggregation function δ to obtain one unified vector per element:

$$\hat{\mathbf{h}}_i = \delta([\hat{\mathbf{h}}_i^0, \hat{\mathbf{h}}_i^1, \cdots, \hat{\mathbf{h}}_i^m]) \quad (8)$$

which are further used by different downstream tasks.

C. Learning Constraint with Design Principle

One characteristic of document design is that layouts usually follow certain design principles[2]. For example, the principle of contrast encourages to distinguish elements of different roles by creating strong visual differences (e.g., color, size). This provides us an ideal signal to constrain the layout representations. Hence, we encode the principles into a loss term which is adapted from the contrastive learning [12]:

$$\mathcal{L}_{aux} = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} |\frac{1}{|K|}\sum_{k\in K} \mathrm{sim}(\mathrm{f}_{\mathrm{i}}^{\mathrm{k}}, \mathrm{f}_{\mathrm{j}}^{\mathrm{k}}) - \mathrm{sim}(\hat{\mathbf{h}}_{\mathrm{i}}, \hat{\mathbf{h}}_{\mathrm{j}})| \quad (9)$$

where K is a heuristically chosen subset of contrastive features. The goal is to constrain the representations of two elements to be closer when they share similar features. $\mathrm{sim}(\mathrm{f}_{\mathrm{i}}^{\mathrm{k}}, \mathrm{f}_{\mathrm{j}}^{\mathrm{k}})$ is the k-th feature similarity between two elements, which is measured by Euclidean distance for normalized numerical features and Jaccard similarity for categorical ones:

$$\mathrm{sim}(\mathrm{f}_{\mathrm{i}}^{\mathrm{k}}, \mathrm{f}_{\mathrm{j}}^{\mathrm{k}}) = \begin{cases} 1/\sqrt{(f_i^k - f_j^k)^2}, & \text{numerical} \\ |f_i^k \cap f_j^k|/|f_i^k \cup f_j^k|, & \text{categorical} \end{cases} \quad (10)$$

$\mathrm{sim}(\hat{\mathbf{h}}_{\mathrm{i}}, \hat{\mathbf{h}}_{\mathrm{j}})$ is the similarity between the output states of two elements, for which we use the inverse of the mean absolute distance.

Finally, we add this auxiliary loss term in the downstream task training (take the cross entropy loss for the classification tasks as an example here) with a hyper-parameter γ to control the loss weight:

$$\mathcal{L} = \sum_{i=1}^{n} -\log P(\hat{y}_i|\mathbf{x}_i;\theta) + \gamma\mathcal{L}_{aux}. \quad (11)$$

here $P(\hat{y}_i|\mathbf{x}_i;\theta)$ is the pseudo likelihood of our model prediction, where $\hat{y}_i$ is the true label for the i-th element, θ is the model parameters and $\mathbf{x}_i$ denotes for the i-th input element.

IV. Experiments

In this section, we give more details of our experimental settings, and evaluate the effectiveness of our disentangled attention mechanism on two datasets. We also present an example downstream application of Layout Suggestion in the supplementary material.

Feature	Description
Element ID	Unique ID with ascending paragraph order.
Alignment Type	Alignment type of the paragraph, including *Left, Right, Center, Justified*.
Indentation	Indentation value of the paragraph.
Font	Font setting such as font type and font size.
Word Count	The number of words in the paragraph.
Text Style	Including four features attached to the text as *Bold/Italic/Underline/AllCaps*. *AllCaps*: indicates whether the words in the paragraph are all capitalized.
Color	RGBA value of the filled color of an element, as well as color gradient, solid, and pattern.
List Structure	Whether the paragraph belongs to an item list or a table cell.
Context Statistics	Such as the number of empty lines before/after the paragraph.

TABLE I: Paragraph features in word document data.

[2] https://en.wikibooks.org/wiki/Graphic_Design/Principles_of_Design

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Model	Macro F1	Recall	Precision	Title	Subtitle	Footer	Heading	Decorator	Caption	Body
XGBoost	81.08	79.45	83.83	98.12	81.14	88.12	64.55	92.80	70.69	72.16
LightGBM	80.70	80.29	83.65	98.93	84.25	94.73	39.26	90.76	78.19	78.81
TabNet	82.56	84.84	80.96	98.63	82.60	90.83	63.61	96.14	76.85	69.23
CanvasEmb	87.00	88.77	86.89	99.14	89.87	**96.74**	69.25	97.84	81.28	74.87
-w/ constrained loss	87.86	89.45	86.92	**99.23**	89.24	96.63	69.76	98.04	80.15	81.95
DeAttnFormer (Ours)	88.29	89.40	**88.22**	99.03	86.38	95.43	75.58	98.11	**82.63**	80.85
-w/ constrained loss	**88.70**	**90.89**	86.90	98.93	**89.95**	95.50	**76.28**	**98.54**	77.06	**84.65**

TABLE II: Element role labeling results for slide data.

Model	Macro F1	Recall	Precision	Title	Subtitle	Heading	SubHeading	BodyText	ListItem
XGBoost	66.11	67.05	66.80	76.41	29.69	54.82	54.54	89.37	91.85
LightGBM	67.07	64.97	71.01	77.21	25.64	59.66	56.14	89.57	94.17
TabNet	61.36	63.60	60.54	69.69	22.36	48.30	46.38	89.23	92.20
CanvasEmb	76.75	82.04	73.71	84.63	59.48	65.75	**64.50**	91.26	94.90
-w/ constrained loss	77.95	81.39	75.45	84.83	67.68	68.14	62.44	90.24	94.41
DeAttnFormer (Ours)	78.91	79.78	**78.27**	**87.57**	68.26	68.19	62.80	**91.94**	94.69
-w/ constrained loss	**79.60**	**82.54**	77.28	86.75	**69.15**	**70.82**	64.07	91.79	**95.00**

TABLE III: Element role labeling results for word document data.

Model	Macro F1	AP	AUC
XGBoost	81.65	90.31	97.49
+ shape relations	85.37	**98.36**	92.84
LightGBM	83.90	98.35	93.37
+ shape relations	85.93	93.18	98.28
TabNet	79.34	96.86	88.00
+ shape relations	83.03	89.94	97.48
CanvasEmb	94.16	94.60	97.83
-w/ constrained loss	95.29	96.09	98.32
DeAttnFormer (Ours)	**95.59**	96.60	**98.70**
-w/ constrained loss	95.39	96.43	98.63

TABLE IV: Image captioning results for slide data.

A. Datasets

We verify our model on two datasets: slide data proposed in CanvasEmb, and word document data created in this paper. Both the datasets contain rich layout information and annotations of the two tasks introduced in Section II-B.

- **Slide Data**. CanvasEmb contains two annotated datasets. For the task of element role labeling, they annotate 48,309 slides (80%/20% for train/test), which classifies each element in a slide into 7 roles: *title* (21.28%), *subtitle* (1.24%), *footer* (1.44%), *decorator* (0.44%), *heading* (0.20%), *caption* (1.23%), and *other* (72.32%). For image captioning task, they annotate 2,996 slides, each with at least one image-captioning pair.
- **Word Document Data**. We construct a dataset of 2,609 word documents (.docx) of various types (e.g., new articles, forms, resumes) crawled from the web. In average there are 210 paragraphs per document. Each paragraph contains multiple designed-related features, with descriptions for each feature listed in Table I. Each paragraph is annotated by 5 labelers for the task of element role labeling, and the final role is assigned via majority vote. The label distribution is: *title* (1.28%), *subtitle* (0.32%), *heading* (7.30%), *subheading* (8.58%), *body* (38.79%), *listItem* (44.0%). We also divide the documents into 80%/20% for train/test.

Though several multi-modality datasets such as FUNSD [13] and DocBank [14] also contain layout information (defined as the work token bounding box), their tasks heavily rely on the text content instead of the layout information. We do not include these datasets in this paper since they require special considerations to the text/image modality such as large-scale pre-training to achieve state-of-the-art performances.

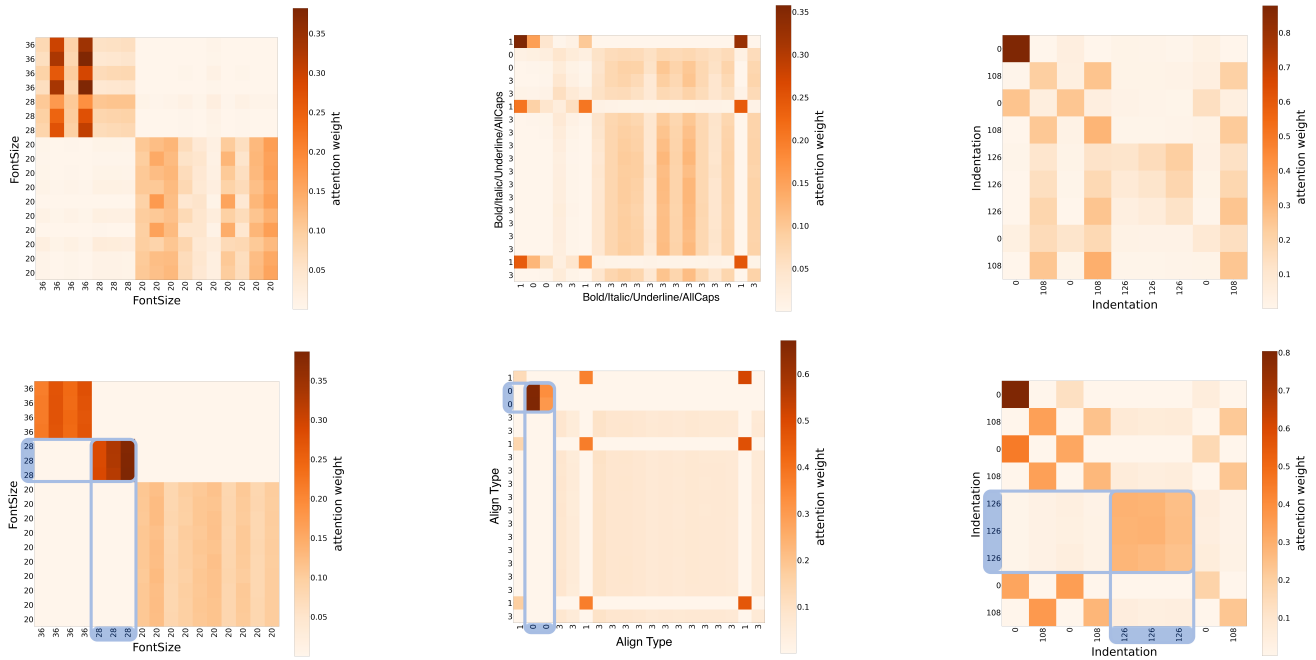
B. Model Implementation Details

We set the Transformer with 6 layers of hidden size 64. The number of heads for intra-feature attention is set to 8. The maximum sequence length is 16 in slide and 128 in word document. For a word document, we split it into chunks if the number of paragraphs exceeds the maximum length. Following CanvasEmb, we use Adam optimizer [15] with initial learning rate of 5e-4, $\beta_1 = 0.9$, $\beta_1 = 0.999$ and l_2 weight decay of 5e-4. The learning rate is adaptively decayed following a linear schedule. We also apply early stopping, gradient clipping [16], and warmup over the initial 1% training iterations with the rate 2e-3. The dropout rate is set to 0.3. Meanwhile, we use the α-balanced cross-entropy loss to deal with label imbalance, where α_c for class c is calculated as $1/\mathrm{proportion(c)}$. The choosing of contrastive feature set and hyper-parameter γ to control the loss weight in Section III-C is based on grid search which we will discuss later in the ablation of constrained loss.

C. Baselines

- **XGBoost and LightGBM**. Decision trees are natural for processing heterogeneous features. We use two state-of-the-art models: XGBoost [17] and LightGBM [18].

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(a) Font Size (b) Bold/Italic/Underline/AllCaps (c) Indentation

Fig. 4: Element-to-Element attention visualizations of three feature channels. The x and y-axis are the corresponding feature values of paragraphs. They are from CanvasEmb (first row) and DeAttnFormer intra-feature attention (second row) from the same document.

Label encoding is adopted for preprocessing categorical features.

- **TabNet [19]**. TabNet is a tabular data learning model that implements decision tree with sequential feature selections and reasoning using Transformer.
- **CanvasEmb [7]**. CanvasEmb is a Transformer-based model which is described in Section III-A. In this paper we do not leverage its pre-training as to only focus on the model architecture, thus to eliminate possible variances brought by different pre-trainings schemes.

Hyper-parameter choose. For GDBT models and TabNet, we apply a simple grid search for optimal parameters. And we adopt the same training and testing scheme for Transformer-based models. Parameters sizes of the neural-based models are controlled to be in the same scale.

Other baselines. Recently there are several models for document representation such as LayoutLM and DocFormer. We do not include them as baselines since they mainly focus on multi-modality integration with extra text and image input, and their models share the same Transformer backbone as CanvasEmb but with different pre-training data and schemes.

D. Results

Here we show the overall results of the two tasks.

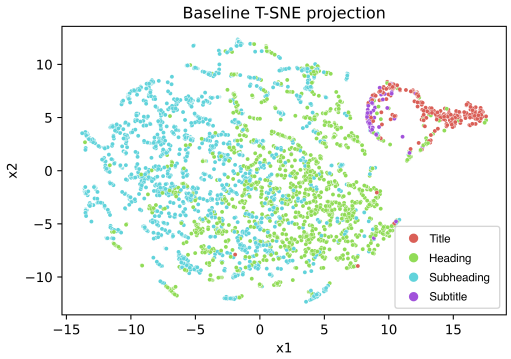
Element Role Labeling. We report the macro Precision/Recall/F1 as well as F1 per class. The results for slide data are shown in Table II and word document data in Table III. We can see that Transformer-based models achieve more competitive results than traditional decision trees. Our model DeAttnFormer surpasses all the baselines in most categories by a large margin. When equipped with the constrained loss, the performances are further boosted in both CanvasEmb and DeAttnFormer, especially on the dominant class *body* (slide) and *listItem* (word document), which proves its effectiveness and generalization capability. In all, our DeAttnFormer achieves the best macro F1 of 88.70% (slide) and 79.60% (word document).

Image Captioning. We report the macro F1, Average Precision and AUC in Table IV. As this task depends on element pair relations, we explicitly add pre-defined relation features (e.g., relative distance) to the decision tree and TabNet baselines, which is shown to boost these baselines. As we can see, our model achieves the best scores in Macro F1. With the constrained loss, CanvasEmb is further improved while DeAttnFormer performs comparably. One possible reason might be that the disentanglement in DeAttnFormer already learns pairwise element relations with respect to informative features (e.g., position), which serves similar goal as the constrained loss. We will show more evidences later in Section V-A.

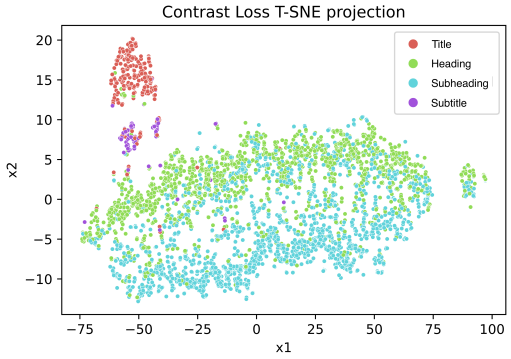
V. Discussions

In this section, we conduct extensive model analyses. We use the word document data for demonstration.

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(a) DeAttnFormer



(b) w/ constrained loss

Fig. 5: T-SNE plot of the output representations of four element roles on word document data.

A. *Effect of Intra-Feature Attention*

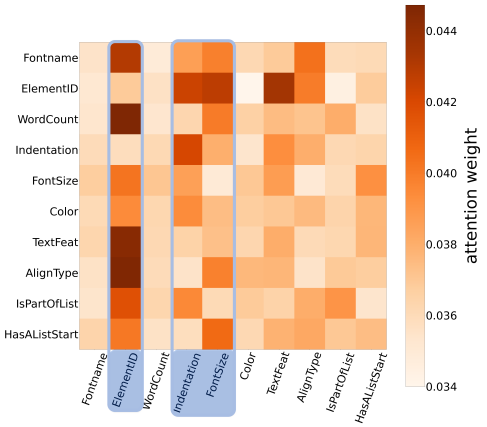
When applying the intra-feature attention, element embeddings per feature channel are separately updated, as to avoid undesired correlations between features. To validate this, we select three features and visualize the heatmaps of both CanvasEmb self-attention (first row) and our intra-feature attention (second row) for comparison in Fig. 4. Highlighted in blue, we observe that our approach produces more reasonable patterns. For example when looking at `FontSize`, in CanvasEmb, paragraphs of font size 28 have high attention weights to the paragraphs of font size 36, but low attention weight to same font paragraphs. While being compared, the attention in our approach is more explanatory that paragraphs with the same font size attend to each other more closely.

B. *Effect of Inter-Feature Attention*

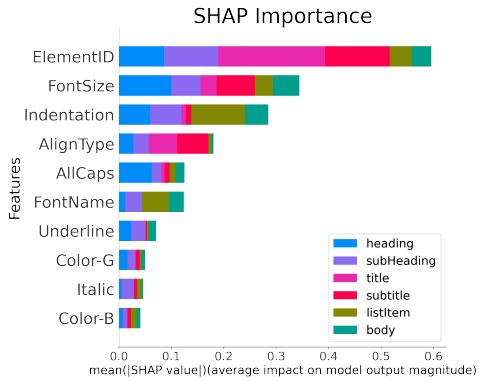
Inter-feature attention aims to capture feature correlations. To check what it has learnt, we extract the inter-feature attention matrices from 40 randomly sampled documents, and visualize their average weights in Fig. 6a. By column-wisely looking at the heatmap, there exist impactful features such as `ElementID` and `Indentation`, to which most of the other features have high attention weights. This is well-aligned with SHAP feature importance [20] in LightGBM in Fig. 6b.

β in Equation 6	Macro F1	Recall	Precision
(1) $\mathbb{1}$ (w/o inter-feature attention)	75.49	79.58	72.54
(2) $1/m$ (averaging weights)	76.62	80.58	73.80
(3) DeAttnFormer (learnable)	**79.60**	**82.54**	**77.28**

TABLE V: Ablations of different inter-feature attention strategies on word document element role labeling.



(a) inter-feature attention



(b) SHAP feature importance in LightGBM

Fig. 6: (a) DeAttnFormer inter-feature attention heatmap, indicating feature-to-feature influences. By column-wisely looking, the most impactful features (highlighted in blue) are similar to the top-ranked important features in (b).

It demonstrates the possibility of DeAttnFormer to implicitly learn feature importance via inter-feature attention.

Furthermore, we show results in Table V using different strategies for obtaining inter-feature attention weights β in Equation 6. As we can see, our model performs the best compared to the models without inter-feature attention (using identity matrix $\mathbb{1}$, no interaction between feature channels) or with simple averaging weights.

C. *Effect of the Constrained Loss*

The auxiliary loss in Section III-C constrains the hidden representations of elements via design principles. We use simple

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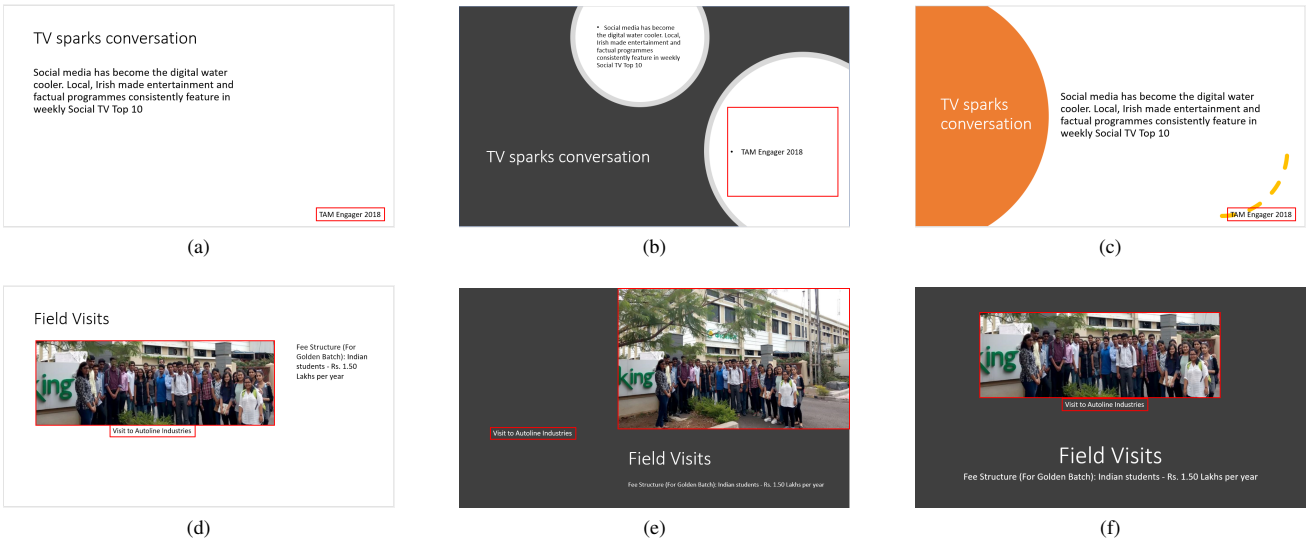


Fig. 7: Two examples of layout suggestion. Given (a)(d) query slide, (b)(e) are the layout candidates by default, and (c)(f) are candidates after applying element role labeling and image captioning, which improves the suggestion quality.

grid search for the choice of the contrastive feature set and the weight γ in Equation 11. Specifically, we select top-5 features from SHAP feature importance score by LightGBM (shown in Fig. 6(b)), and for each feature, we try $\gamma \in [0.5, 0.1, 0.01]$ and choose the setting with the best performance on dev. The best contrastive feature on word document data is `ElementID` with $\gamma = 0.5$, while on slide data is `Position` with $\gamma = 0.1$. The observation suggests that constrains from spatial-related features are favored.

To better illustrate the effect of the constrained loss, we visualize the learned element representations on word document data in Fig. 5 using t-SNE [21]. We select the four roles *title*/*subtitle* and *heading*/*subheading* as they are the most ambiguous pairs. As we can see, the model exhibits more discernible clustering of different roles in the projected space after applying the constrained loss.

VI. Application: Layout Suggestion

Systems of layout suggestion can be divided into template-based and generation-based. We follow the template-based approach since it is more controllable and stable which is the current mainstream choice in many commercial products. Template-based systems mainly consist of three sub-modules: (1) layout template retrieval given a user query; (2) candidate generation by mapping query elements to template placeholders; (3) candidate ranking to present the final suggestion list to users[3].

Layout understanding on the query slides, such as element role labeling and image captioning, helps better template retrieval and candidate generation. We illustrate the details by showing two examples in Fig. 7. Different from well-defined templates, query slides usually ignore semantic roles of elements as well as grouping information. For example in Fig. 7a, when adding a footnote (highlighted in the bottom-right corner), most users would directly insert a textbox without specifying its semantic role as a footnote. In this case, it is likely to retrieve the wrong template and map the footnote to a main bodytext as shown in Fig. 7b, while detecting the footnote can solve this issue (Fig. 7c). As another example in Fig. 7d, the highlighted text caption is a textbox without explicitly grouping it with the image as a whole. Thus Fig. 7e wrongly separates the two elements into different placeholders, and Fig. 7f provides a more accurate mapping with the output of image captioning. The above two cases indicate that a better representation along with accurate layout understanding can increase the quality of layout suggestions.

[3]The ranking function depends on different features such as user preference and template popularity. We do not focus on the sub-module in this case.

VII. Conclusion

When applying Transformer in document layout representation, simple summation of heterogeneous feature embeddings could bring mixed correlations in the self-attention. In this paper, we present a novel two-step disentanglement (DeAttnFormer) to alleviate this issue. We first disentangle the attentions per feature channel with intra-feature attention, then model feature interactions with inter-feature attention. Further, we propose an unsupervised learning objective to constrain layout representations, which encodes the document design principles and requires no annotation effort. Experiments on two layout understanding tasks prove the effectiveness and generalization of our approach. Extensive analyses further demonstrate that our approach can produce more explainable attention patterns with more reasonable feature interactions.

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